Sequences, Series, Limits & Continuity

A *sequence* of real numbers is a function, u, mapping $\mathbb{N}$ to $\mathbb{R}$

- We will be dealing with infinite sequences
- Written $u(1), u(2), u(3), ...$, or $(u_1, u_2, u_3, ...)$
- The sequence is written $\{u_n\}$
- Example of a sequence

$$\frac{3n^2}{n^3+9}$$

which gives the sequence

$$\frac{3}{10}, \frac{12}{17}, \frac{48}{73}, ...$$

Arithmetic sequences

- An arithmetic sequence is a sequence $\{u_n\}$ in which

$$u_{n+1} - u_n$$

is constant for all n

- Equivalent to saying that for some numbers a, d, the sequence looks like

$$a, a + d, a + 2d, a + 3d, ...$$

or

$$u_n = a + (n - 1)d$$

- Is this sequence arithmetic? (If so, what are a, d?)

$$u_n = 14 - 5n$$

- Is this one?

$$u_n = 5^{-n}$$

Arithmetic sequences: Each term is obtained by adding same number to previous term

Geometric sequences: Each term is obtained by multiplying previous term by same number

- Equivalent to saying that for some numbers a, x the sequence looks like

$$a, ax, ax^2, ax^3, \ldots$$

or

$$u_n = ax^{n-1}$$

- What are a, x for this geometric sequence?

$$u_n = 5^{-n}$$

Limits of sequences

- The *limit* of a sequence describes what happens to u_n as n becomes large
- A sequence $\{u_n\}$ tends to the limit z if for *any* positive real number ϵ there is a positive integer, N, such that:

$$|u_n - z| < \epsilon \text{ for all } n > N.$$

- Also written as

$$\lim u_n = z$$

$$\lim_{n\to\infty} u_n = z$$

$$u_n \to z \text{ as } n \to \infty$$

An example: Let $u_n = \frac{1}{n}$

Convergent sequence: $\lim u_n = z$ if $|u_n - z| < \epsilon$ for all $n > N$.

Take any $\epsilon > 0$ (it can be arbitrarily small). Let N be any integer greater than $\frac{1}{\epsilon}$. Note that $N > \frac{1}{\epsilon}$ implies that $\epsilon > \frac{1}{N}$.

Then for all $n > N$, we have

$$|u_n - 0| = \frac{1}{n} < \frac{1}{N} < \epsilon.$$

It follows that our sequence $u_n = \frac{1}{n}$ tends to the limit $z = 0$.

Definitions:

A sequence that tends to a limit is called *convergent*

A sequence that tends to infinity (or negative infinity) is *divergent*, but a sequence doesn't have to tend to infinity to be divergent (what's an example of a divergent sequence that doesn't tend to infinity?)

A *Cauchy* sequence of real numbers is a sequence whose elements cluster around some number as n gets large. Since we are dealing with the real numbers, any convergent sequence is Cauchy, and vice versa.

$\{x_n\}$ is Cauchy if $\forall \epsilon > 0, \exists N \in \mathbb{N} : |x_m - x_n| < \epsilon$ when $m, n > N$

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We can show the sequence defined by $x_n = n$ is not convergent:

$|x_n - x_m| = |n - m| \geq 1$ for all n, m. Thus, for $\epsilon = 1$ there is no N such that $|x_n - x_m| < \epsilon$ whenever $n, m > N$. Therefore $\{x_n\}$ is not Cauchy, and does not tend to a limit.

Working with limits (there are many more identities in Chapter 31)

Fact 1: For a real number x, $\lim_{n\to\infty} x^n = 0$ if and only if $-1 < x < 1$.

Fact 2: For a real number b, $\lim_{n\to\infty} n^b = 0$ if and only if $b < 0$.

What's the limit of

$$u_n = \frac{n+1}{2n}$$

Claim: If $\lim u_n = z$ then $\lim |u_n| = |z|$

Proof: If $\lim u_n = z$ then for any $\epsilon > 0$, $|u_n - z| < \epsilon$ for n large. First, note that

$$||a| - |b|| \leq |a - b|$$

(the "reverse triangle inequality"). This basically proves our result: For any $\epsilon > 0$ we have $|u_n - z| < \epsilon$ for n large, and consequently $||u_n| - |z|| < |u_n - z| < \epsilon$. Thus, $\lim |u_n| = |z|$. $\square$

Does the converse hold? (If $\lim |u_n| = |z|$ then $\lim u_n = z$)